

Nukes and Ladders: India and Pakistan's Nuclear Evolution

Abstract

Recent modernization initiatives by both India and Pakistan within the last two decades have outpaced the growth of their command and control capacities. As both countries diversify conventional and nuclear arsenals, neither have mitigated the inherent risks of their technological evolutions, both in strategy and operationalization. Furthermore, India has worked to “reset” its conservative military reputation as Pakistan continues its support for non-state Islamist actors in Kashmir. This paper argues that the reactionary nature of India and Pakistan’s military advancements and strategic doctrines is inherently escalatory. Conventional arguments assert that robust nuclear deterrence in South Asia ensures stability. However, this paper argues that the ongoing arms race has heightened the risk of conflict by increasing “use it or lose it” pressures and introducing newer, riskier technologies. Particularly, this paper finds that India’s increased conventional counterforce capacity and Pakistan’s growing non-strategic capabilities have created new avenues for conflict. An analysis of India and Pakistan’s military technology reveals that while advancements may seem to prevent large-scale conflict, they also destabilize the region and escalate the potential for nuclear confrontation.

Introduction

In February 2019, India and Pakistan were on the brink of war. On February 14th, a suicide bomber attacked a Central Reserve Police Force convoy near Pulwama, Kashmir. The blast, claimed by banned Pakistan-based terrorist organization Jaish-e-Mohammed (JeM), killed 40 military personnel. Two weeks later, Narendra Modi ordered an airstrike to eliminate a JeM terrorist camp in Balakot. For the first time since 1971, Indian aircraft carried out an airstrike on Pakistani territory. The next day, the Pakistani Air Force launched a strike in Jammu in retaliation. In the ensuing engagement, India allegedly shot down a Pakistani F-16 while losing one of their own MIG-21, along with its pilot who was forced to eject over Pakistani territory and was subsequently captured¹. Modi’s decision was unprecedented—a striking contrast to a history of self-restraint. Even during the height of the Kargil War, the Indian Air Force was strictly restrained from crossing the Line of Control that delineates Indian and Pakistani-controlled Kashmir.

The Pakistani response itself was alarming. Although the PAF bombing raid was purely conventional, Islamabad convened a meeting of its nuclear command body—the National Command Authority. Notably, the NCA also convenes to discuss conventional military responses. However, to Indian officials, this was a veiled nuclear signal—Pakistan seemed intent on linking the crisis with nuclear escalation to encourage India to deescalate. It is unclear how the crisis would have deescalated had Wing Commander Varthaman not been captured by

¹ <https://www.stimson.org/2022/three-years-after-balakot-reckoning-with-two-claims-of-victory/>

Pakistani forces. After their aerial strike and the capture of WC Varthaman, Modi reportedly considered a missile strike should Pakistan not return the Indian pilot. Modi also claimed to deploy nuclear-powered missile submarines, readying India's nuclear arsenal². In the aftermath of Balakot, Modi has repeatedly emphasized the veracity of his threat, asserting he was ready to commit a "night of murder."³ Pakistani officials also claim the threat was real. Should India have launched their missiles, then-Prime Minister Imran Khan warned Pakistan would respond "three times over"⁴. Without the pilot as a lynchpin for diplomacy—a chance occurrence—there were several avenues for intense escalation.

Before either country could escalate further, quiet bilateral mediation facilitated Pakistan's release India's pilot, resulting in a gesture of peace at the Wagah border. The crisis eventually de-escalated. Nonetheless, there remains the sentiment that the Balakot crisis resolved itself mostly by chance. Pakistan's reprisal and convening of the NCA was similarly unprecedented as India's use of airpower across the LOC.⁵ Both India and Pakistan repeatedly miscalculate the other nation's resolve, with theories of de-escalation that seem incompatible. As India leaves strategic restraint behind, the world cannot rely solely on luck to de-escalate crises between two nuclear-armed states.

In this paper, I argue that the reactionary nature of India and Pakistan's military and technological developments are escalatory. Contrary to conventional arguments, while nuclear deterrence may seem to prevent large-scale conflict, the action-reaction dynamic of the nuclear arms race that inadvertently ensnares India and Pakistan has increased so-called "use it or lose it" pressures, increased the scope of limited nuclear exchange, and introduced newer, riskier military technologies that neither country are entirely prepared to use responsibly. Indeed, Balakot, rather than solely demonstrating how conflicts can be solved conventionally, also highlights how quickly and unpredictably conflicts can escalate. It is difficult to forget, for both nations, how rapidly India and Pakistan readied their nuclear forces—whether by convening the NCA or flushing out nuclear submarines. The potential for crisis instability is only bolstered by the nations' advancing nuclear capabilities. As such, an analysis of current and future technological developments in India and Pakistan reveal how arms competition continues to destabilize South Asia and escalate the risk of nuclear conflict.

Many conventional arguments assert that India and Pakistan has entered a period of greater stability largely because of their technological advancements and the ever-present threat of nuclear escalation. India's strategic restraint was regarded as a way to avoid breaching

² <https://www.stimson.org/2020/from-kargil-to-balakot-southern-asian-crisis-dynamics-and-future-trajectories/>

³ "PM Shri Narendra Modi Addresses Public Meeting in Patan, Gujarat: 21.04.2019," *Bharatiya Janata Party*, YouTube video filmed and posted April 2019, <https://www.youtube.com/watch?v=hxZGbwCcCd0>.

⁴ Sanjeev Miglani and Drazen Jorgic, "India, Pakistan Threatened to Unleash Missiles at Each Other: sources," *Reuters*, 16 March 2019, www.reuters.com.

⁵ Yusuf, Moeed W. "The Pulwama Crisis: Flirting With War in a Nuclear Environment." *Arms Control Today* 49, no. 4 (2019): 6–11. <https://www.jstor.org/stable/26732415>.

Pakistan's nuclear threshold. As Sumit Ganguly argues in *Nuclear Stability in Nuclear South Asia*, "an examination of the onset, evolution, and termination of the 1999 and 2001-02 crises between India and Pakistan suggest that nuclear deterrence is robust in South Asia"⁶. Ganguly echoed this sentiment in March of 2019, two weeks after Pakistan returned India's pilot. He asserted that historical precedence proved that neither country is eager to cross the adversary's nuclear threshold; instead, because of the potential catastrophic consequences, conventional methods of conflict-resolution are heavily relied upon⁷. That Pakistan convened their chain of nuclear command chain and still resorted to a conventional response—despite India's clear escalation—was further evidence that nuclear deterrence stabilized the region. Ashley J. Tellis similarly expressed in *Striking Asymmetries: Nuclear Transitions in South Asia* that for now, nuclear deterrence is robust in the region. He finds "high degrees of nuclear deterrence and crisis stability despite the persistent possibilities of Indian conventional military action provoked by Pakistan's nuclear-shadowed campaign of terrorism against India"⁸. Observers assert that because India and Pakistan have been on the brink of conflict many times before but still haven't escalated, nuclear deterrence is still effective.

However, despite this supposed security, it is undeniable that India and Pakistan have undergone distinct military modernizations. Over the last two decades, India and Pakistan have expanded their nuclear and conventional arsenals. Notably, India has increased its intelligence and offensive capabilities—both in conventional and nuclear options, alluding to a potential shift towards counterforce targeting. Pakistan has worked to assure its second-strike survivability—at the very least, to be able to assure retaliation—while acquiring tactical nuclear weapons. Both countries seem caught in an action-reaction dynamic. Barry Buzan describes the action-reaction model in his *Military Technology and International Relations*. The model theorizes that "states will arm themselves either to seek security against threats posed by others, or to increase their power to achieve political objectives against the interests of others."⁹ Any action taken by either state to increase its military strength also raises the level of threat the state poses to other states.¹⁰ To avoid insecurity, the adversarial state perceives the necessity to increase their own strength. This model was often utilized to illustrate Cold War dynamics between the United States and the USSR, particularly to explain their nuclear and qualitative conventional arms races. India's military modernization—whether targeted at Pakistan or simply to increase its global standing—has increased Pakistani concerns that India is moving away from the posture of minimum

⁶ Ganguly, Sumit. "Nuclear Stability in South Asia." *International Security*, 2008.

⁷ Ganguly, Sumit. "Why the India-Pakistan Crisis Isn't Likely to Turn Nuclear." *Foreign Affairs*, March 5, 2019. <https://www.foreignaffairs.com/articles/india/2019-03-05/why-india-pakistan-crisis-isnt-likely-turn-nuclear>.

⁸ Tellis, Ashley J. "Striking Asymmetries: Nuclear Transitions in Southern Asia." *Carnegie Endowment for International Peace*, July 2022. <https://carnegieendowment.org/research/2022/07/striking-asymmetries-nuclear-transitions-in-southern-asia>.

¹⁰ Buzan, Barry. "Chapter 6." In *Introduction to Strategic Studies: Military Technology and International Relations*. London: Palgrave Macmillan, 1987. Pg. 76.

deterrence. Simultaneously, Pakistan's nuclear development forces India to anxiously reckon with Pakistan's evolving nuclear threat. Arguably, India and Pakistan's concerns and accompanying military build-up mirror the tenuous Cold War balance between the U.S. and USSR.

Within the action-reaction process, there are three main variables: *magnitude*, the proportionality of the reaction to the triggering action, *timing*, the sequence and speed of the interaction, and *awareness*, the degree to which both states know of their impact on the other ex-ante.¹¹ In the case of India and Pakistan, who are significantly concerned with the threat of the other, both states are aware of how they impact the other—as evidenced by the frequent references made by each government to the military developments of the other. Furthermore, it can be argued that several shifts in thinking—particularly India's Cold Start doctrine in the early 2000s and Pakistan's tactical arsenal—are developed in direct reaction to a “triggering event”—the 2001 Parliament Attack and the development of Cold Start respectively.¹² Such a dynamic adequately summarizes India and Pakistan's peacetime military advancements.

However, in the process of force development, one must also contend with how states *anticipate* their adversaries' perception and behaviors in crisis bargaining and decision-making. Since it often takes up to ten to fifteen years for a state to design, build, and deploy a weapon, states must react to their predictions of what other states will do¹³. A missile could take little more than 7 minutes to reach a significant target. Due to the quick turnover of strategic decisions between India and Pakistan, leaders must anticipate how the other state will act. Such interactions are often termed *spiral model*: two actors “locked into a smooth, continuous, and self-reinforcing pattern of mutual military stimulation.”¹⁴ Indeed, while Pakistan and India's military developments can often be read as direct reactions to each other, there is an anticipatory nature to how they plan their future developments, intent on gaining an advantage on the other.

India

A Reset in Reputation

India and Pakistan's relationship is only complicated by the various pathways for conflict between the two states, India's recent reputational reset, and the impact of China on the Indo-Pak dyad. Conflict between the two states is most notably linked to militant non-state actors. Pakistan has often exploited Islamist non-state actors as a tool to confront India, allowing Pakistan to shape the strategic environment of South Asia without reckoning with India's stronger

¹¹ Ibid.

¹² Ladwig, Walter C. III. "A Cold Start for Hot Wars? The Indian Army's New Limited War Doctrine." *International Security* 32, no. 3 (2008): 158-190.

¹³ Glaser, Charles L. "When Are Arms Races Dangerous? Rational versus Suboptimal Arming." *International Security* 28, no. 4 (Spring 2004): 44-84. <https://doi.org/10.1162/0162288041588313>.

¹⁴ Jervis, Robert. "Chapter 2." In *The Illogic of American Nuclear Strategy*, 15-38. Ithaca: Cornell University Press, 1974.

conventional forces. Such policy emboldens Pakistan to continuously challenge Indian control of Kashmir.¹⁵ Even if Pakistani forces do not explicitly or directly support non-state Islamist actors, Indian officials, notably External Affairs Minister S. Jaishankar, nonetheless publicly state their belief in Pakistan's sponsorship of terrorism¹⁶. This strategy, while once holding India at a stalemate, has become a liability to Pakistani interests—to combat non-state actors, India has developed more efficient offensive options and has strived to “reset” its reputation from one of strategic restraint to one of precise action.

India responded to the 2001 Parliamentary attack with Operation Parakram—a full-scale mobilization that intended to thrust into Pakistan’s plains and desert sectors, one of the most vulnerable targets in the Pakistani homeland. However, India never fully executed Parakram. Not only was the mobilization too slow to act before the U.S. intervened, but Pakistan also readied its nuclear forces and explicitly threatened to use nuclear options in response to any decisive conventional attack. Parakram ended with an Indian retreat, having not achieved even its minimum objectives.¹⁷ Reeling from Parakram’s failure, Indian defense strategists devised the Cold Start doctrine: quicker mobilizations and conventional operations calibrated to be right under Pakistan’s nuclear threshold. Cold Start has three objectives:

- “Capture a strip of territory of Pakistan and retain it as a post-conflict bargaining chip,
- Limit the incursions of Pakistani forces into Indian territory,
- Avoid triggering a Pakistani nuclear response.”¹⁸

Through Cold Start, India intended to work around Pakistan’s nuclear threat while still being able to conventionally respond to Pakistani provocation. Pakistan, however, decidedly preempted Cold Start by Pakistan’s development of battlefield, non-strategic weapons that lowered Pakistan’s nuclear threshold and allowed for limited nuclear use.¹⁹

Prior to 2014, India has often implemented strategic restraint to avoid crossing Pakistan’s nuclear threshold. From delaying sending forces to defend Kashmir in 1947 to avoiding testing nuclear weapons until 1998, the nation’s history of military force has been conservative. Most notably, India’s self-imposed caveat that they would not enter Pakistan-occupied Kashmir during the Kargil crisis—despite refusing any diplomatic overtures until all of the invaders were

¹⁵ Kapur, S. Paul, and Sumit Ganguly. “The Jihad Paradox: Pakistan and Islamist Militancy in South Asia.” *International Security* 37, no. 1 (2012): 111–41. <http://www.jstor.org/stable/23280406>.

¹⁶ International, Asian News. “‘there Will Be Consequences’: S Jaishankar on Pakistan Practising Terrorism.” NDTV.com, May 17, 2024. <https://www.ndtv.com/india-news/there-will-be-consequences-s-jaishankar-on-pakistan-practising-terrorism-5683859>.

¹⁷ Narang, Vipin. *Posturing for Peace? Pakistan’s Nuclear Postures and South Asian Stability*. Cambridge: Harvard University Press, 2022.

¹⁸ Bukhari, Syed Adnan Athar. “Tactical Nuclear Weapons and Deterrence Stability in South Asia.” Edited by Sam Dudin and Chelsey Wiley. *The 2019 UK PONI Papers*. Royal United Services Institute (RUSI), 2019. <http://www.jstor.org/stable/resrep37409.5>.

¹⁹ Kapur, S. Paul. “Ten Years of Instability in a Nuclear South Asia.” *International Security* 33, no. 2 (Fall 2008): 71–94.

evicted—solidified Pakistan’s belief that India feared escalation and was willing to sacrifice military efficiency for long-term stability. India’s measured response, in the end, bolstered its international reputation as a responsible country who sought stability. However, domestically, both the armed forces and the citizenry were disconcerted by India’s inability to respond to Pakistan’s nuclear linkage, a sentiment that only grew stronger after the 2001 terrorist attacks on Parliament and the Mumbai 2008 terror attack.²⁰

Over the last decade, India has made significant efforts to mitigate the stranglehold imposed by Pakistan's nuclear threat on its military response. In 2016, in response to a terrorist attack on an army outpost in Uri which killed 19 soldiers, the Indian government ordered a surgical strike into POK against militant launch pads. While the events of the Indian strike are greatly disputed, Indian officials claimed 35-50 casualties, a fact that Indian media corroborated. The strike itself seemed incredibly popular, garnering public and political support. Before the next election cycle, Bollywood even produced the movie, *Uri: The Surgical Strike*, which became the tenth highest-grossing film by March 5th, 2019²¹. EAM Jaishankar asserts that the 2014 election—when the BJP was elected to power—was a clear decision that the people of India would no longer tolerate any cross-border terrorism. He condemned India’s early passiveness towards “put[ting] up with terrorism as an eccentricity of our neighbor.” He pointed to this public approval when defending “the message of Uri and Balakot”²². In contrast to earlier administrations—who had tempered their response to terrorist attacks—the current BJP government made it clear that India will no longer restrain herself.

Whether this shift in risk attitude is internally or externally motivated, it has undoubtedly contributed to changes in the strategic environment of India and Pakistan by opening new pathways for conflict. It is entirely possible, as we saw in Balakot, that India may retaliate conventionally to terrorist attacks, prompting Pakistan to react in kind. Such a shift, coupled with India’s robust modernization, particularly increased precision and ISR capabilities, and Pakistan’s nuclear diversification, which could allow for limited nuclear strikes to combat India’s conventional superiority, could increase the likelihood of violence.

China’s Shadow on India-Pakistan Relations

China is the world’s fastest growing nuclear power. For the last three years, Beijing has added 100 new nuclear warheads to its arsenal every year. A 2023 Pentagon report asserts that China is outpacing previous projections of its nuclear build-up, capable of acquiring over 1000 warheads by 2030. A 2022 military report also claims that China could possess 1500 warheads

²⁰ Cheema, JS, (retired). 2024. “The enduring lesson of strategic restraint.” The Tribune India. May 9, 2024. Accessed July 29, 2024. <https://www.tribuneindia.com/news/comment/the-enduring-lesson-of-strategic-restraint-619295>.

²¹ “'Uri' box office collection Day 53: The Vicky Kaushal starrer military drama is now the tenth highest grosser film ever”. The Times of India. 5 March 2019.

²² “Dr S. Jaishankar on 'Western' Criticism of India's Ukraine Stand,” YouTube video, 15:32, posted by NDTV, March 1, 2023, <https://www.youtube.com/watch?v=QypiuZ4ZgXM>.

by 2035 as President Xi looks to meet and surpass the U.S.'s global standing.²³ Although China has reassured the world that it will not move away from its "no first use" posture, senior U.S. military officials argue that China may not see NFU as applicable in certain conflict scenarios.²⁴ As tensions ratchet over Taiwan, China's force structure and strategy are morphing to become more aggressive. China is increasing the diplomatic, military, and economic pressures it can leverage on, not just Taiwan, but all of its regional neighbors as it seeks to combat U.S. efforts to contain China's rise.

Undoubtedly, China influences India's decision-making and weapons development strategies. The India-Pakistan-China triangle is colored by the conventional gap between India and Pakistan and the nuclear gap between India and China. The India-China dyad features mitigating factors such as economic cooperation, the lack of sub-nuclear military conflicts, confidence building measures, and civilian control over NC3 which induce relative stability despite nuclear asymmetries. Nonetheless, when India conducted its nuclear test in 1998, it cited China's threat as a reason for its weapons development. China has considered India as a target of a potential nuclear attack since the 1980s.²⁵

Particularly recently, as India attempts to compete with China on the global stage, New Delhi and Beijing's relationship has grown more confrontational. In 2020, a skirmish at the Ladakh outpost on border resulted in the first combat casualties in 45 years.²⁶ Clashes broke out in the Galwan valley, heightening tensions higher than they had been since 1965. EAM Jaishankar referred to this clash when explaining why the relationship with China has yet to recover. Until China respects the Line of Control, relationships are unlikely to improve under the current administration. While China's force buildup is primarily directed at matching U.S. global hegemony, it only worsens the strategic asymmetry between India and China. India does not have the resources nor the intent to try and quantitatively match China's nuclear build-up. India's military modernization is partially aimed at reducing the depth of asymmetry between its arsenal and China's.²⁷ Chinese nuclear development over the next twenty years will prompt Indian decision-makers to adapt to the changing strategic environment; while the nuclear asymmetry between the two countries has yet to truly destabilize the region, as China seeks new nuclear

²³ U.S. Department of Defense. Military and Security Developments Involving the People's Republic of China 2023. October 19, 2023. Accessed July 31, 2024. <https://media.defense.gov/2023/Oct/19/2003323409/-1/-1/1/2023-MILITARY-AND-SECURITY-DEVELOPMENTS-INVOLVING-THE-PEOPLES-REPUBLIC-OF-CHINA.PDF>.

²⁴ Knickmeyer, Ellen. "Pentagon Report: China on Track to Triple Its Nuclear Warheads by 2030." *Associated Press*, October 19, 2023. Accessed July 25, 2024. <https://apnews.com/article/china-military-nuclear-missiles-pentagon-taiwan-ukraine-3040f2dd6d02c63924e94e5a873cfb1b>.

²⁵ Kurita-China-India-Relations

²⁶ Peri, Dinakar. "India, China Tensions Continued through 2021 and 2022, New Details Reveal." *The Hindu*, January 17, 2024. <https://www.thehindu.com/news/national/india-china-tensions-continued-through-2021-and-2022-new-details-reveal/article67746294.ece>.

²⁷ Adil, Ahmad. "Relations with China Can Only Be Better If It Respects Line of Actual Control: India." *Anadolu Agency*, July 29, 2024. <https://www.aa.com.tr/en/asia-pacific/relations-with-china-can-only-be-better-if-it-respects-line-of-actual-control-india/3288659>.

heights, India may be forced to dramatically alter how it conceives its nuclear options to deal with the greater threat.

Both India and China have nuclear postures of assured retaliation—securing credible deterrence against nuclear attacks and ensuring survivable nuclear forces to execute retaliation even after a strike by an adversary. However, in terms of survivability, India's second-strike capability is still weak. In this context, its development of long-range and Multiple Independently Targetable Reentry Vehicles (MIRV) missiles focused on maintaining second strike survivability and credible deterrence against China. On March 11, 2024, India successfully launched the first test flight of its Agni-V MIRV missile off of Odisha on its eastern coast. With a range of more than 5,000 kilometers and its ability to hit multiple locations, it can effectively target major China cities far better than any of India's other missiles. The Agni-V, due to its range, was already referred to as the "China Killer." With added MIRV capabilities, India has grown closer to truly credible deterrence against China.²⁸

Furthermore, India's nuclear development is another method of competing with China for global prestige. Unlike its military confrontations with Pakistan, shadowed by nuclear threats and shows of force, India does not yet seem to take China's nuclear threat seriously. Retired Vice Admiral of the Indian Navy, Verghese Koithara, notes that although nuclear deterrence against China is a topic of India's domestic security discourse, there has not been true serious and practical thought towards challenges of ensuring second strike survivability.²⁹ Rajesh Basrur, in *India and China: Nuclear Rivalry in the Making?* further argues that India strives to attain second strike survivability out of principle rather than due to any imminent threat that China, who also has a No First Use posture, poses to the Indian homeland³⁰. Indeed, observers note that India's MIRV capabilities lets the nation join an elite club of nuclear states³¹. In the context of India's general military modernization, the Agni-V not only lets India level its competition with China but projects India's determinations to be part of the global elite.

Nonetheless, while India competes with China, Pakistan observes from India's other border. By trying to challenge its asymmetry with China, India disturbs the nuclear balance it has with Pakistan. India's March 11th test of Agni-V drew Islamabad's criticisms and accusations that India had violated an agreement on advance notification of ballistic missile tests.³² Foreign Office Spokesperson Mumtaz Zahra Baloch sees the Agni-V as an extension of India's new

²⁸ Roy, Rajesh. "India Adds Firepower to a Missile Program Focused On ..." India Adds Firepower to a Missile Program Focused on China, March 11, 2024. <https://www.wsj.com/world/asia/india-adds-firepower-to-a-missile-program-focused-on-china-564dcd0e>.

²⁹ Kurita, pg. 52

³⁰ Rajesh Basrur, *India and China: Nuclear Rivalry in the Making?* (Singapore: S. Rajaratnam School of International Studies, October 2013), p. 7, https://www.rsis.edu.sg/wp-content/uploads/2014/07/PB131001_India_and_China_Nuclear_Rivalry.pdf.

³¹ Joshi, Yogesh. "Strategic Shifts: India's MIRV Milestone and Nuclear Policy Dynamics." *The Diplomat*. March 2024. <https://thediplomat.com/2024/03/strategic-shifts-indias-mirv-milestone-and-nuclear-policy-dynamics/>.

³² Yousaf, Kamran. "Pakistan Accuses India of Violating Accord with Agni-5 Test." *The Express Tribune*, March 14, 2024. <https://tribune.com.pk/story/2459351/pakistan-accuses-india-of-violating-accord-with-agni-5>.

aggression and the BJP's hawkishness. She referenced concerns, in the same statement, over India's implementation of the controversial Citizenship Amendment Act and the government's alleged Hindutva philosophy. Pakistan also allegedly possesses MIRV capabilities with its Ababeel missile—however, experts are skeptical about whether the MIRV technology is functional. India's successful test is worrying.³³ The Agni-V is much more than a technological milestone; it is a missile that can reach any target on the Pakistan map. In response to India's achievement, Pakistan must consider how it can counter India's growing nuclear threat—especially given the threat MIRV technology can pose to nations without sophisticated ballistic missile defense systems like Pakistan. Indeed, India's need to compete with China only bolsters the South Asian arms race. Every threat aimed at China will also be heard by Pakistan.

Military Modernization: Becoming a Global Military Power

A hallmark of Modi's BJP India has been defense modernization and reform. India's military has been partly motivated by the threat of China's military, which is ahead in terms of sustained military expenditure and modernization. As a result, India's military has worked to cope with the changing contours of modern conflict and the shifting geopolitical stage. Moreover, India is often behind on military innovation and over reliant on external weapons providers—most notably Russia. To counter this, India's aim is to indigenize its defense research and development while catching up to the global military elite. This has included initiatives to reorganize command structure, encourage private sector defense involvement, and introduce new, technological advances into India's arsenal.³⁴ Sections of this paper will discuss India's advancements in intelligence-surveillance-reconnaissance (ISR) capabilities, Ballistic Missile Defense (BMD), nuclear weapons delivery systems, and conventional projectiles. India is also working to increase road and transportation infrastructure.

India has accelerated its defense design and manufacturing through involving new external entities and incentivizing investment and expenditure. Where the public sector once dominated defense projects, the Modi government encouraged significant private sector involvement. Simultaneously, India incorporated more French, Israeli, and American manufactured components to lessen its dependence on Russian-made equipment, coupled with domestic platforms, design, and development. Such a model also involves secure cooperation with other military powers. This includes cooperative agreements such as one that allows India and the U.S. to share significant ISR capabilities and joint maritime surveillance operations with Australia³⁵. India shares very close military relationships with the U.S., Russia, France, Israel,

³³ "Ababeel." *Missile Threat: CSIS Missile Defense Project*. Center for Strategic and International Studies. Accessed July 29, 2024. <https://missilethreat.csis.org/missile/ababeel/>.

³⁴ Singh, Mandeep. "India's Defense Evolution Targets Modernization, Reform, Global Engagement." *Indo-Pacific Defense Forum*, January 7, 2024. <https://ipdefenseforum.com/2024/01/indias-defense-evolution-targets-modernization-reform-global-engagement/>.

³⁵ Ladwig, Walter C., and Anit Mukherjee. "Understanding India's Military Modernisation." *RSIS*, January 24, 2024. <https://www.rsis.edu.sg/rsis-publication/idss/ip24024-understanding-indias-military-modernisation/>.

and Germany, due to, in part, its goal of modernization and its desire to attain poly-alignment with different states.

Such reforms are not without their challenges. The Indian government has long been anti-military, focusing on how to ensure civilian control over the military rather than on how to build an effective fighting force. Civil-military relations were riddled with frustration from the armed forces about financial deficiencies and a lack of influence on policy and decision-making. To combat this, Modi government has made broad strides in integrating the different armed forces and ensuring cohesion and communication between various branches. By creating the new Department of Military Affairs, Modi intends to give the military a legitimate voice in decision-making.³⁶ However, improving organizational structure is not sufficient to overhaul an outdated and disjointed military. While India's defense budget reflects changed priorities in security strategy, India faces resource constraints that prevent full-scale rapid modernization. The budget has steadily increased over the last four years, particularly after the 2020 Galwan valley conflict with China.³⁷ Yet, India's defense budget allocation is still far below the demand of 2.5 to 4 percent of the GDP³⁸. In comparison to the U.S. and China, its defense spending is miniscule.

Modi's reforms are further hindered by inadequate budget allocation, poor political oversight, and a lack of effective legislation to coordinate and manage reforms. Even within the steadily rising defense budget, there are gaps in how the money is spent. As the Stimson center pointed out in their analysis of India's military spending, the amount spent on "capability building" for joint organizations has been minimal, hovering right above one percent of the Joint Staff Budget³⁹. If the Indian military truly desires 'jointness,' they must be willing to invest in it. Furthermore, financial commitments are varied and erratic, continuing the trend of uneven flows and politicization that has haunted Indian defense spending since Independence. In July 2024 itself, the national government has dropped defense spending to focus on jobs and income in the wake of the BJP government's election set-back.⁴⁰

In order to effectively modernize its military, India must also rethink how its forces are used and employed. PM Modi reemphasizes jointness—the idea that Indian forces must think and act cohesively. Furthermore, many scholars suggest that unless India adapts to its rapidly changing strategic environment, its forces risk irrelevancy. Delhi was, for a large part of its history, reliant on an orthodox offensive doctrine, centering on "large army formations, operating relatively autonomously from political direction, seeking to impose a punitive cost on the

³⁶ <https://www.economist.com/asia/2023/11/29/narendra-modi-is-remaking-indias-14m-strong-military>

³⁷ <https://economictimes.indiatimes.com/news/defence/union-budget-2024-from-galwan-to-2024-indias-defence-budget-reflects-strategic-changes/articleshow/111773915.cms?from=mdr>

³⁸ <https://www.stimson.org/2024/indias-military-modernization-efforts-under-prime-minister-modi/>

³⁹ Ibid.

⁴⁰ <https://www.bloomberg.com/news/articles/2024-07-23/india-defense-budget-drops-as-government-focuses-on-jobs>

enemy.”⁴¹ Their nuclear doctrine was similarly unchanging since India’s acquisition of nuclear weapons: a “No First Use” policy that promised “unacceptable damages” should Indian forces be attacked by nuclear forces or other weapons of mass destruction (chemical or biological weapons in the hands of a non-nuclear nation). India’s nuclear thinking has revolved around countervalue targeting. India’s deterrence posture threatens civilian targets should an adversary attack first. As India’s military thinking evolves, this posture has been called into question by military officials, scholars, and politicians.

India’s No First Use: The Counterforce Debate

Vipin Narang and Christopher Clarey, in “India’s Counterforce Temptations: Strategic Dilemmas, Doctrine, and Capabilities,” assert that India is “developing a suite of capabilities and increasingly making statements about preemption and counterforce that appear inconsistent with its professed strategy of assured retaliation or minimum deterrence”⁴². Indeed, as a result of India’s broader military modernization, it has acquired the necessary technology to consider and even enact counterforce attacks—that is strikes on military targets—against Pakistani command, control, communications, and delivery systems. Furthermore, as India strives to counteract Pakistan’s range of non-strategic nuclear options, counterforce targeting is an attractive option.

To enact counterforce targeting, India would need to move away from its NFU doctrine. While there are several reasons for why India may never shift from NFU—the advantages of its reputation as a ‘responsible’ nuclear state, for example—such a change has been discussed by several key officials. In his 2014 writings, Lt. Gen. B.S. Nagal (ret.) who is a former strategic forces commander and an advisor to the Prime Minister on nuclear issues, advocated that India abandon NFU in favor of ambiguity. He believed that India should allow for potential first uses in preemption scenarios. His argument was that NFU, in a democratic state such as India, was amoral. If Indian leadership had knowledge that Pakistan was preparing to use of nuclear weapons, it would be unconscionable for decision-makers to not act in defense of the Indian people⁴³. Lt. Gen. Nagal is not alone in his argument; in 2016, Defense Minister Manohar Parrikar claimed that India should not declare if it has a NFU policy or not. He asserted that unpredictability was central to any strategy⁴⁴. In 2019, Defense Minister Rajnath Singh stated that “till today, our nuclear policy is ‘no first use’. What happens in the future depends on the circumstances”⁴⁵. Whether or not India directly moves away from NFU, statements by such

⁴¹ Tarapore, Arzan. *Ground Forces in India's Military Strategy: Optimizing Their Employment*. Carnegie Endowment for International Peace, 2020. https://carnegie-production-assets.s3.amazonaws.com/static/files/Tarapore_Ground_Forces_in_Indian_Military.pdf.

⁴² Narang, Vipin, and Christopher Clary. "India's Counterforce Temptations: Strategic Dilemmas, Doctrine, and Capabilities." *International Security* 43, no. 3 (Winter 2018/19): 7-52. <https://ssp.mit.edu/news/2019/indias-counterforce-temptations>.

⁴³ Lt. Gen. B.S. Nagal, “Checks and Balances,” *Force*, June 2014.

⁴⁴ Narang and Clarey, “India’s Counterforce Temptations”

⁴⁵ Toby, Dalton. “Much Ado about India’s No-First-Use Nuke Policy.” Carnegie Endowment for International Peace, September 26, 2019. <https://carnegieendowment.org/posts/2019/09/much-ado-about-indias-no-first-use-nuke-policy?lang=en>.

personnel about abandoning it has done little to assure Pakistan that India will abide by its doctrinal promises. Indeed, in a direct conflict, Pakistan may not take India's NFU into account at all when planning its next step or predicting India's next move. Moreover, as India's capabilities advance, its NFU policy seems less credible to Pakistani decision-makers.

Counterforce targeting would quickly open avenues for India to tackle Pakistan's use of sub-strategic warfare and the threat of its non-strategic nuclear weapons. In contrast to India, Pakistan does not have a NFU policy; it reserves the right to leverage its nuclear arsenal to counter India's conventional superiority.⁴⁶ As Pakistan continues to develop battlefield nuclear options, India must contend with how to respond to a sub-strategic attack. A countervalue target may be seen as invalid retaliation and invite further attacks on the homeland; a lack of response will allow Pakistan to take advantage of India's crippled conventional forces. In such a situation, a conventional counterforce attack that can target Pakistan's launch capabilities undoubtedly seems attractive. Again, Pakistan's perception of India's potential counterforce temptations is more impactful than India's true internal shift. While India's growing capabilities may simply be the result of its broader modernization—undoubtedly impacted by global technological trends that prioritize precision and information fidelity—it still signals to Pakistan that India may be trying to develop counterforce targeting abilities. Pakistan, in a crisis, will face greater “use it or lose it” pressures; if India truly possesses the strength and intent to decapacitate Pakistan's launch capability, then Pakistan loses significant battlefield advantage by not launching non-strategic nuclear weapons.

However, counterforce is still technically and tactically very difficult. India would need to accurately find and target the location of Pakistan's nuclear delivery systems. It would need to have precise intelligence and precise delivery systems. Limited counterforce itself is strategically flawed—an attack against one nuclear stockpile does not negate an attack by another. In this scenario, India would also benefit from being able to strike several areas at once. It would also need to strike fast to preempt Pakistani decision-makers who are already wrestling with aforementioned “use it or lose it” pressures. Indeed, India's recent strides in military technology may indicate that India is working towards overcoming these challenges.

ISR—India's Evolving Information Network

The effectiveness of counterforce strategies relies on the ability to successfully complete the kill-chain. In order to conduct a successful strike, an operator would need to:

- “1) Find the target;
- 2) Determine target's location, course and speed;
- 3) Communicate that information coherently to the platform launching the weapon; and,

⁴⁶ Narang, Vipin, 2022.

4) Launch the attack.”⁴⁷

The most challenging steps are accurately locating the target, determining how a system is moving (especially if it is a mobile delivery system), communicating quickly enough that the information is delivered in real-time, and launching an attack that is precise enough to destroy or significantly degrade the target. In order to actualize counterforce targeting, a state must be confident that it can locate and hit at least a majority of the adversary’s delivery systems. If significant capabilities remain, the adversary is still able to launch a significant counterstrike and inflict damage. Counterforce campaigns can focus on damage limitation—that is a large reduction of the adversary’s arsenal to avoid assured destruction. However, Pakistan has over 170 warheads. If Pakistan deploys a sizeable of its warheads, India will still need to destroy a high number of targets.⁴⁸ Locating precise targets requires the ability to obtain reliable real-time information which includes geospatial intelligence with readily available imagery. Therefore, ISR is an essential enabling technology for counterforce targeting. Although it is unclear whether India is developing ISR capabilities in order to operationalize a counterforce strategy, India’s technological strides in ISR have enabled new types of targeting that were formerly unfeasible.

In recent decades, ISR capabilities have become a cornerstone of modern warfare. To effectively plan and execute military operations, conventional or nuclear, real-time or near-real-time information acquired from ISR platforms is vital. As military powers compete for ISR advantages, India has worked to expand its cross-domain ISR abilities to maximize military effectiveness. Furthermore, India is haunted by how suboptimal intelligence has hampered its past response to the Kargil crises.⁴⁹ As a part of its modernization initiative, India is transforming its ISR force structure into a “modern and robust strategic warning system which consists of information gathering satellites, an expanding fleet of airborne platforms, and ground-based sensors.”⁵⁰ As India markets itself as an important actor in restraining Chinese superiority, several western military powers—including the U.S.—have offered assistance in bolstering India’s ISR to ensure adequate surveillance over Chinese forces. As a result, however, Pakistan must contend with India’s increasingly adept information acquisition.

The Indian Space Research Organization (ISRO) has further developed space systems with ground-based Space Situational Awareness and space-based Command, Control, Communications, Computers, ISR (C4ISR) abilities. Furthermore, India is preparing to build heavier class satellites, working towards “dedicated military satellite networks” that could anchor its electronic intelligence, geospatial intelligence, communication, early warning, and navigation

⁴⁷ <https://web.archive.org/web/20130613233413/http://cno.navylive.dodlive.mil/2013/04/23/kill-chain-approach-4/>

⁴⁸ Glaser, Charles L., and Steve Fetter. "Should the United States Reject MAD? Damage Limitation and U.S. Nuclear Strategy toward China." *International Security* 41, no. 1 (Summer 2016): 49-98. https://doi.org/10.1162/ISEC_a_00248.

⁴⁹ Brookings Institution. "20 Years after Kargil: Military Operations, Perceptions, and Decision-Making." Event. Washington, DC, September 24, 2019. <https://www.brookings.edu/events/20-years-after-kargil-military-operations-perceptions-and-decision-making-2/>.

⁵⁰ Mahmood, Ahmed, and Sultan, Muhammad. "Impact of India’s ISR Capabilities on South Asian Security Dynamics." *Strategic Studies*, January 15, 2023.

systems—this would only augment India's ambitious space strategy. Such developments have worsened the strategic imbalance between Pakistan and India as India cements its superiority in space.⁵¹ To effectively manage the exponential increase in information inputs, India must ensure robust and secure efficient data fusion at headquarters, and reliable downstream bandwidth from headquarters to field weapons. Information management has become a priority in the Army's modernization. As a part of the Army's modernization plans, information management has become a priority. Organizationally, India intends on establishing a network-centric organization and secure communication networks through implementing a Force Multiplier Command Post (FMCP). Such a post would receive and process information at a central and secure location before disseminating it outwards. India's navy has also set up a Maritime Domain Awareness (MDA) system that includes an integrated network of sensors—India plans to expand this system to include surface, subsurface, air, and cyber dimensions.⁵²

The Indian Navy also possesses U.S. manufactured P-8Is, an ISR aerial platform, which contain multiple lasers capable of performing long-range maritime reconnaissance and anti-submarine warfare. It further utilizes the Ilyushin II-38, a Soviet-designed maritime patrol and anti-submarine warfare aircraft. Other aircraft include the anti-submarine warfare Dornier 228 MPA and the reconnaissance Tupolev Tu-142MKE.⁵³ Therefore, the Indian Navy can employ both short-range and long-range maritime patrol with significant anti-submarine capabilities. As Pakistan endeavors to build its sea-based nuclear-leg, it will need to compete with India's increasing maritime intelligence capabilities and anti-submarine measures. To expand both land and maritime intelligence, India is heavily investing in procuring more than 5,000 unmanned aerial vehicles while developing indigenous unmanned *Nishant*, *Rustom-I/II*, and *Netra* systems to implement tactical and medium-altitude-long-range capabilities and bolster aerial-based ISR.⁵⁴ *Nishant* is a multi-mission, day/night, battlefield ISR UAV that improves India's short-range surveillance capabilities. It is intended for artillery fire correction, it can help India overcome uncertainties in situational awareness. Such a UAV could help quickly assess pre-strike movements and post-strike damage.⁵⁵ The *Rustom* UAVs are intended for medium-altitude long-endurance once deployed—*Rustom II* will have strike capability.⁵⁶ *Netra*, developed in collaboration with private company IdeaForge for both military and civilian use, is effective for anti-terrorist and counter-insurgency operations, particularly in dense forested or difficult to access areas.

⁵¹ Ibid.

⁵² Ibid.

⁵³ Singh, Abhijit. "Unmanned and Autonomous Vehicles and Future Maritime Operations in Littoral Asia." *Observer Research Foundation*, February 2021. <https://www.orfonline.org/research/unmanned-and-autonomous-vehicles-and-future-maritime-operations-in-littoral-asia/>.

⁵⁴ Mahood and Muhammed, 2023.

⁵⁵ Defence Research and Development Organisation. "Unmanned Aerial Systems (UAS)." Ministry of Defence, Government of India. Accessed August 5, 2024. <https://www.drdo.gov.in/drdo/unmanned-aerial-systems-uas>.

⁵⁶ "Rustom MALE UAV." Integrated Dynamics Pakistan. Last modified November 28, 2012. Accessed August 5, 2024. <https://web.archive.org/web/20121128070736/http://idp.justthe80.com/unmanned-aerial-systems/rustom-male-uav#TOC-Rustom-H-Capabilities>.

Beyond indigenous ISR advancements, India intends to increase inter-operability and military communication between itself and the U.S. Through the Communications Compatibility and Security Agreement (COMCASA), India has access to real-time operational information from U.S. space-based assets as well as sensitive communication equipment. These assets allow India to leverage shared intelligence and real-time imagery to improve situational awareness on the battlefield. In addition, India signed the Basic Exchange and Cooperation Agreement (BECA) with the U.S. which allows for information sharing on maps, satellite imagery, military-grade coordinates, and other critical data. The actual effects of these agreements are uncertain. The U.S. is unlikely to share information that would enable an Indian counterforce strike. However, Pakistani decision-makers will consider the partnership's potential applications in an Indo-Pak conflict. Through real-time access to US geospatial intelligence, India can improve the accuracy of its missiles and air-launched weapons—including its recently acquired S-400 surface-to-air-missile defense system. Particularly in the context of counterforce targeting, BECA may provide India the necessary information to precisely target an adversary while COMCASA can help India react quickly to a changing battlefield scenario.⁵⁷ Pakistan is undoubtedly very anxious about U.S.-India cooperation; Pakistani defense planners have repeatedly discussed concerns about Indian preemptive strikes, particularly in collaboration with an outside power like the U.S. and Israel.⁵⁸ Regardless of India's true motivation, Pakistan will perceive Delhi's security and ISR agreements as a threat to its nuclear capabilities.

Regardless of India's ISR achievements, its capabilities are still far below what they would need to be to effectively conduct a counterforce strike. States take significant steps to strengthen, conceal, and protect their delivery systems. While Pakistan does not have a missile defense system, the nation has presumably taken measures to ensure survivability. Details are intentionally sparse to maintain operational security, but Pakistan likely employs mobile systems, hardened and deeply buried facilities, camouflage and concealment, and information compartmentalization to protect against a preemptive strike against its nuclear systems.⁵⁹ Furthermore, many of Pakistan's short-range and medium-range missiles are road-mobile, including its non-strategic Nasr-class missiles. In his 2010 article on challenges to U.S. nuclear superiority, James Acton asserts that "it is still fiendishly difficult to locate mobile missiles hidden by a well prepared enemy."⁶⁰ Against road-mobile weapons, even the U.S. could not eliminate an entire state's arsenal with high probability. In comparison, India's ISR is markedly inferior. If Pakistan is on high alert, already weary of potential preemptive strikes, then they could maneuver road-mobile systems to avoid low-yield strikes.

⁵⁷ Ibid.

⁵⁸ Lavoy, Peter R. "Pakistan's Nuclear Posture: Security and Survivability." Monterey: Naval Postgraduate School, Center for Contemporary Conflict, 2007. Accessed August 5, 2024.
https://www.npolicy.org/article_file/Pakistans_Nuclear_Posture-Security_and_Survivability.pdf.

⁵⁹ Ibid.

⁶⁰ James M. Acton, 'Managing Vulnerability,' *Foreign Affairs* 89/2 (2010), 147.

Conventional Weapons: Improving Accuracy

Counterforce targeting does not necessitate nuclear weapons; with enough intelligence about the target's location and the extent of hardening, conventional weapons are enough to inflict sufficient damage. India's continued mission to bolster indigenous precision-guided missile production seems to have given India the lead in its missile race with Pakistan. While the conventional missile asymmetry is currently manageable for Pakistan decision-makers, India's future strides and continued investment threatens to leave Pakistan behind. Furthermore, India has worked to make its missiles lighter, harder to intercept, and more precise, thus increasing fears of preemptive strikes.

Cruise missiles are characteristic counterforce weapons. They feature high speed and precision, can be launched from multiple launch platforms, and possess stealth features that can bypass an adversary's defenses.⁶¹ India's BrahMos supersonic cruise missiles, originally intended to be sea-based anti-ship missiles, were adapted to land, sea, and air platforms and deployed to the Indian Navy in 2005 and the Army in 2007.⁶² The current BrahMos-I has a range of around 300 kilometers. However, India is working with Russia to develop the hypersonic BrahMos-NG (Next Generation) missile. BrahMos-NG is projected to have the same range while being significantly lighter than its predecessor. Its smaller size would enable deployment on many platforms in greater numbers than the BrahMos-I, bringing the missile closer to the battlefield.⁶³ India and Russia are also jointly developing the BrahMos-II, a hypersonic scramjet-propelled missile with a projected range of 1,500 km and speed of Mach 8.⁶⁴ As BrahMos Aerospace reports, the BrahMos-II will be a "brilliant weapon against deeply buried enemy nuclear bunkers and heavily protected locations." The project is currently stalled as Russia negotiates its commitment, but developers are optimistic about its continuation. Another of India's assets is Nirbhay, India's first indigenously produced cruise missile. It utilizes a land-based mobile launcher; however, reports imply that it can potentially be deployed on submarines. The development of Nirbhay, BrahMos-NG, and BrahMos-II indicate that India is undoubtedly exploring hypersonic options that are difficult to defend and effective against hardened targets.

At the onset of India's Cold Start doctrine, cruise missiles were used to bolster India's doctrinal shift. Cold Start envisioned limited, precise strikes. Nirbhay, with its extended range, is capable of crossing the border into Pakistan; colloquially called the "backbone of cold start," Nirbhay was seemingly developed to ensure a 'flexible response' to Pakistan's battlefield nuclear

⁶¹ Kulsoom Belal. "Cruise Missiles in South Asia: Implications for Regional Strategic Stability." *Policy Perspectives* 13, no. 1 (2016): 115–34. <https://doi.org/10.13169/polipers.13.1.0115>.

⁶² Jalil, Ghazala Yasmin. "Missile Race in South Asia: Security Challenges for Pakistan in the 21st Century." *Strategic Studies* 40, no. 1 (2020): 39–57. <https://www.jstor.org/stable/48732326>.

⁶³ Sputnik News. "Hypersonic Variant of BrahMos in the Pipeline: CEO." Last modified June 12, 2024. <https://sputniknews.in/20240612/hypersonic-variant-of-brahmos-in-the-pipeline-ceo-7597184.html>.

⁶⁴ BrahMos Aerospace. "India to Develop BrahMos-II Missile." Last modified August 3, 2009. <https://brahmos.com/newscenter.php?newsid=95>.

weapons.⁶⁵ Even as India moves away from Cold Start, Pakistani leaders will be paranoid about how such weapons may be deployed into the Pakistani homeland. Further, India's intention to modify the BrahMos missile so that it can be deployed from the Su-30MKI brings India closer to a "devastating deep strike weapon."⁶⁶ The Su-30MKI, incredibly deadly in its own right, is a dramatic force multiplier when paired with the supersonic BrahMos. Even without a nuclear warhead, a conventionally armed BrahMos could penetrate hardened NC3 targets. The Su-30MKI, the most capable fourth generation fighter, is adept at penetrating hardened air defenses, greatly increasing the chances of a pilot reaching a designated target.⁶⁷ Indeed, India has also worked to acquire and indigenously develop aircraft that can intrude deep inside enemy territory for effective delivery. Apart from the Su30s, India also has the Vajra (Mirage-2000H), Bhadur (Mig-27), Mig-27, Jaguar IS, and Dassault Rafale aircraft—all potentially capable of delivering both conventional and nuclear warheads in Pakistani territory. While information about the nuclear role of these aircraft is vague, India has striven to adapt precise weaponry to powerful and stealthy aircraft.⁶⁸

Beyond its cruise missiles, India possesses several surface-to-surface missiles with varying ranges and speeds. Since 2000, India developed 11 missiles, including the short-range Prithvi and Prahaar and the medium and intermediate-range Agni ballistic missiles. The Prahaar ('to strike' in Sanskrit) cruise missile, a "short-range, solid propellant, road-mobile ballistic missile designed for tactical strikes against close range targets."⁶⁹ India has carefully not posited a nuclear payload for Prahaar, but it could ostensibly be used as a tactical nuclear weapon, often seen as a potential competitor to Pakistan's *Nasr* missile.⁷⁰ Its newer derivative, Pralay, draws elements from both Prahaar as well as the K-series of submarine-launched ballistic missiles and the Prithvi Defense Vehicle from the Ballistic Missile Defense Program.⁷¹ Many of India's ballistic missiles are suspected to be dual-capable. Some ballistic missiles may be nuclear-only, particularly the longer-range Agni missiles which are less precise than their short-range counterparts. However, Pakistan will likely consider how India can utilize ballistic missiles like Prahaar and Pralay on the battlefield, both in conventional and nuclear warfare.

⁶⁵ The Hindu. "Nirbhay Will Be Backbone of Cold Start, Say Experts." Last modified October 24, 2014. <https://www.thehindu.com/news/national/nirbhay-will-be-backbone-of-coldstart-say-experts/article6529087.ece>.

⁶⁶ Simha, Rakesh Krishnan. "Why the BrahMos Armed Sukhoi Is Bad News for India's Enemies." Russia Beyond, April 20, 2015. https://www.rbth.com/blogs/2015/04/20/why_the_brahmos_armed_sukhoi_is_bad_news_for_indias_enemies_42687.

⁶⁷ Ibid.

⁶⁸ Kristensen, Hans M., and Matt Korda. "Nuclear Notebook: How Many Nuclear Weapons Does India Have in 2022?" Bulletin of the Atomic Scientists, July 19, 2022. <https://thebulletin.org/premium/2022-07/nuclear-notebook-how-many-nuclear-weapons-does-india-have-in-2022/>.

⁶⁹ Khan, Feroz Hassan. "Pakistan's Tactical Nuclear Weapons and Their Impact on Stability." Carnegie Endowment for International Peace. Last modified June 30, 2016. <https://carnegieendowment.org/research/2016/06/pakistans-tactical-nuclear-weapons-and-their-impact-on-stability?lang=en>.

⁷⁰ Bukhari, Syed Adnan Athar. "Tactical Nuclear Weapons and Deterrence Stability in South Asia." Edited by Sam Dudin and Chelsey Wiley. *The 2019 UK PONI Papers*. Royal United Services Institute (RUSI), 2019. <http://www.jstor.org/stable/resrep37409.5>.

⁷¹ Rout, Hemant Kumar (8 September 2018). "[Pralay set for maiden launch](#)". *The New Indian Express*.

Precise ballistic missiles are incredibly expensive to develop and maintain. Yet India has invested heavily to emphasize a level of accuracy it does not need for countervalue targeting.⁷² While such improvements align with India's ultimate ambition of emerging as a prestigious military power, India's missile developments enable counterforce targeting by providing precision for both short-range and longer-range targets.

India's Ballistic Missile Defense

One of the challenges posed by counterforce targeting is the low probability of completely demolishing the adversary's nuclear capabilities. Should systems survive, a retaliatory strike could still cause damage. The ability to absorb an adversary's remaining arsenal is necessary when considering a full-scale counterforce campaign. Therefore, missile defense systems are enabling technologies for counterforce targeting. Even if some of Pakistan's systems survive, India's quest for a robust ballistic missile defense system indicates India's burgeoning ability to absorb a weakened nuclear attack from Pakistan. As India has dealt with Pakistan's use of nuclear coercion to negate India's conventional superiority, a BMD shield seems like a logical tool for India to place in its arsenal.

Ballistic missile defense systems can be destabilizing for nuclear deterrence. Deterrence is contingent on mutual vulnerability—if one state is notably less vulnerable than its adversary, then deterrence can become unstable. One argument is that India's acquisition of BMD would question the credibility of Pakistan's nuclear deterrence and increase India's resolve.⁷³ The possession of a BMD system, should it test effectively, would allow India to take greater risks by leveraging its protection against potential nuclear escalation. Furthermore, the extreme cost of developing or acquiring a credible BMD system might prevent Pakistan from attaining one of their own, deepening the military asymmetry between the two states. However, other arguments conclude that India's BMD is not yet advanced enough to have a significant effect on South Asian deterrence.⁷⁴ The real capabilities of India's acquired systems are not entirely known; in spite of this, several Pakistani analysts remain vocally confident that its improvements in its missile arsenal, particularly the MIRVed Ababeel missile, can defeat any dedicated Indian anti-missile defense systems.⁷⁵

Nonetheless, India has been working towards trustworthy BMD for over two decades, regardless of Pakistani diplomatic overtures to halt BMD development and maintain nuclear balance. India's current system is a combination of indigenous innovation and acquired elements from its strategic partners (namely Israel, Russia, and the U.S.). The U.S., in particular, is

⁷² Ajey Lele and Parveen Bhardwaj, "India's Nuclear Triad: A Net Assessment," IDSA Occasional Paper, No. 31 (New Delhi: Institute for Defense Studies and Analyses, 2013), pp. 30–31.

⁷³ Jalil, Ghazala Yasmin. "Indian Missile Defence Development: Implications for Deterrence Stability in South Asia." *Strategic Studies* 35, no. 2 (2015): 29–46. <https://www.jstor.org/stable/48527460>.

⁷⁴ Ansari, Usman. "Can Pakistan Counter India's New S-400 Air Defense System?" *Defense News*, August 19, 2022. <https://www.defensenews.com/global/asia-pacific/2022/01/16/can-pakistan-counter-indias-new-s-400-air-defense-system/>.

⁷⁵ Ibid.

motivated by India's intention to combat China's Anti-Access/Area-Denial strategy, which impacts targets in Indian waters, through robust BMD technology that can level some of India's conventional and nuclear asymmetry with China.⁷⁶

India possesses a two-tiered BMD system: the Prithvi Air Defense (PAD) and the Advanced Air Defense (AAD). The PAD system is a high altitude system intended for mid-course interception while the AAD system is a low altitude system intended for terminal stage interception. These systems comprise phase I of Indian BMD development—the interception of missiles with up to 2000 km range. As phase II of its program, India is also developing the AD-1 and AD-2 interceptors to protect against intermediate range and intercontinental ballistic missiles.⁷⁷ India also negotiated a deal to acquire 5 battalions the Russian S-400 surface-to-air defense system⁷⁸, one of the most effective missile defense systems on the market.⁷⁹ Three of the systems have already been delivered with the remaining two to be delivered by the end of 2026.

On June 24, 2024, India tested the second phase of its BMD system—the AD-1 interceptor—which has a reported range of 5,000 km. Indian Defense Minister Rajnath Singh commented that the test “again demonstrated India's ballistic missiles defense capability.”⁸⁰ The test fully met all the objectives, “validating complete network centric warfare weapon system consisting of Long Range Sensors, low latency communication system and MCC (Missile Control Centre) and Advance Interceptor missiles.”⁸¹ India's DRDO tested the AD-1 with a hostile ballistic missile that the system detected and intercepted. Three days later, India tested its S-400 system; according to test reports, it “shot down 80 percent of the ‘enemy’ fighter aircraft package while forcing others to retreat.”⁸² India is also working on creating its own long-range air defense system, Project Kusha, and long-range surface-to-air missile to “detect and neutralize aerial threats such as stealth fighters, aircraft, ballistic and cruise missiles, precision-

⁷⁶ Nagal, Balraj. “India and Ballistic Missile Defense: Furthering a Defensive Deterrent.” Carnegie Endowment for International Peace, June 30, 2016. <https://carnegieendowment.org/research/2016/06/india-and-ballistic-missile-defense-furthering-a-defensive-deterrent?lang=en>.

⁷⁷ Khan, Ghazala Yasmin Jalil. *Emerging Military Technologies and Strategic Stability in South Asia*. Institute of Strategic Studies Islamabad, August 8, 2024. https://issi.org.pk/wpcontent/uploads/2024/08/IB_Ghazala_Aug_8_2024.pdf.

⁷⁸ Center for Strategic and International Studies. “S-400 Triumph.” Missile Threat, last modified August 8, 2024. <https://missilethreat.csis.org/defsys/s-400-triumf/>.

⁷⁹ The Economic Times, “Russian S-400 Triumph gives India the edge against Pakistan, China,” October 17, 2016, <http://economictimes.indiatimes.com/news/defence/russian-s-400-gives-india-an-edge-against-pakistan-china/articleshow/54893457.cms>.

⁸⁰ “India Successfully Tests Phase-II Ballistic Missile Defence System,” Business Standard, July 25, 2024, https://www.business-standard.com/external-affairs-defence-security/news/india-successfully-testsphase-ii-ballistic-missile-defence-system-124072401336_1.html.

⁸¹ Ibid.

⁸² “S-400 air defence system 'shot down' 80% of entire 'enemy' package in exercise,” India Today, July 27, 2024, <https://www.indiatoday.in/india/story/indian-air-force-iaf-s-400-air-defence-system-militaryexercise-russia-2572492-2024-07-27>.

guided munitions, and unmanned aerial vehicles (UAVs) to ranges of nearly 350 km.”⁸³ These tests indicate that Indian decision-makers have some degree of trust in their defense systems. While it is difficult to ascertain whether this degree is high enough for India to risk a retaliatory strike, Pakistan decision-makers, working under a worst-case assumption, might believe their deterrence threat to be decidedly challenged.

Beyond traditional ballistic missile defense systems, India has worked to attain anti-satellite weapons (ASAT) which, while designed to destroy satellites, may also be leveraged against ballistic weapons. According to Ankit Panda, these weapons belong to a broader category of ‘hit-to-kill’ systems that use kinetic force of an object to destroy another.⁸⁴ ASAT weapons are designed to destroy low-orbit satellites; however, the task of destroying a ballistic nuclear-armed missile is very similar. There are certain trade-offs: satellites are 28 times faster but more predictable than ballistic missiles, which, while slower, can appear without much warning.⁸⁵ While Mission Shakti, India’s ASAT development program, is still far from realizing true robust missile defense, India is undoubtedly investing in technologies that could one day enable counterforce targeting.

Increasing Nuclear Readiness

India’s recent nuclear advancements have seemingly prioritized which may be worrisome to Pakistani decision-makers as it adds to India’s unpredictability in a crisis. With its Agni-5 MIRVed missile, renamed the Agni-P, India is embarking on a mission to integrate so-called “next-generation” nuclear capable ballistic missiles into its arsenal. Notably, the Agni-P’s mobile launch system is canisterized. Canisterizing is the process of “storing missiles inside a sealed, climate-controlled tube to protect them from the outside elements during transportation.”⁸⁶ This configuration would allow the warhead to be permanently mated rather than installed prior to launch, decreasing the amount of time necessary to launch nuclear weapons. Not only would this improve India’s readiness but it would also reduce signs of escalation—signals involving mating an uncanisterized missile with its warhead are more visible in a crisis than the relatively seamless launch procedure for a canisterized missile.⁸⁷ Indeed, former senior civilian security officials and SFC officers have implied that “some portion of the nuclear force, particularly those weapons and capabilities designed for retaliation against Pakistan, are now kept at a much higher state of readiness, capable of being operationalized and

⁸³ “India Initiates Long-range Surface-to-air Missile Programme,” Janes, November 6, 2023, <https://www.janes.com/osint-insights/defence-news/defence/india-initiates-long-range-surface-to-airmissile-programme#:~:text=The%20LRSAM%20system%20will%20be,350%20km%2C%20the%20official%20added.>

⁸⁴ Panda, Ankit. “India Can Blow up Satellites Now. A New Space Arms Race Could Be Starting.” *The Washington Post*, April 1, 2019. <https://www.washingtonpost.com/outlook/2019/04/01/india-can-blow-up-satellites-now-new-space-arms-race-could-be-starting/>.

⁸⁵ Ibid.

⁸⁶ Kristensen, Hans M., and Matt Korda. “India’s Nuclear Arsenal Takes a Big Step Forward.” Federation of American Scientists, March 13, 2023. <https://fas.org/publication/indias-nuclear-arsenal-takes-a-big-step-forward/>.

⁸⁷ Ibid.

released within seconds or minutes—not hours, as has been previously assumed.”⁸⁸ India has long held a posture of credible minimum deterrence; yet, Narang notes, the move to increase readiness during peacetime does indicate aggressiveness instead of minimal force.

In relation to counterforce targeting, the inability to ascertain India’s movements from visible signaling will concern Pakistani decision-makers who are already paranoid about a preemptive strike. The ability to surprise the adversary—at least to the degree that significant measures cannot be taken to protect nuclear capabilities—is a necessary precondition for the success of a counterforce strike. As such, as India canisterizes its arsenal, crisis instability may be amplified. If Pakistan cannot know when India will strike, it will be motivated to not only increase its readiness but to act before a potential Indian decapitating attack.⁸⁹ Difficulties regarding a true counterforce strike have already been noted; nonetheless, canisterizing, whether intended for counterforce or not, has shifted Pakistani perceptions of India’s capacity and willingness to launch a preemptive assault.

Pakistan

Non-State Actors in Pakistan

When asked when India and Pakistan would normalize relations, EAM Jaishankar responded that “the ball is in their [Pakistan’s] court.” According to him, India has “struggled continuously with the fact that they have unrelentingly practiced terrorism.”⁹⁰ Especially recently, it is India’s view that frequent cross-border terrorism is not only allowed but supported by Pakistan. S. Paul Kapur argues that Jihad is a central pillar of Pakistan’s grand strategy. Pakistan has long since exploited Islamist non-state actors to confront India sub-conventionally—deliberately avoiding direct military confrontation with India’s superior conventional forces.⁹¹ Pakistan’s intelligence and military have repeatedly relied on such non-state actors to carry out operations, weaken adversaries, and attain their security interests—however, the support of these actors have paradoxically weakened Pakistan’s own security and international standing. Pakistan, of all the N9, faces the highest rate of internal violence by non-state actors, and has been assessed by U.S. analysts to have a high-risk of non-state attacks on nuclear targets. Therefore, the state must not only account for how to protect its nuclear command, control, and stockpiles from potential malicious actors but will also need to reconcile with India’s newfound willingness to escalate in reaction to attacks by non-state actors. So long as Pakistan supports—or is seen as supporting terrorist groups—tensions in South Asia will remain high.

⁸⁸ Narang, Vipin. 2013. “Five Myths about India’s Nuclear Posture.” *The Washington Quarterly* 36 (3): 143–57. doi:10.1080/0163660X.2013.825555.

⁸⁹ Hans M. Kristensen and Matt Korda, “India’s Nuclear Arsenal Takes a Big Step Forward,” *Federation of American Scientists*, March 13, 2023.

⁹⁰ International, Asian News, “‘There Will Be Consequences’: S Jaishankar on Pakistan Practising Terrorism.”

⁹¹ Kapur, S. Paul. *Jihad as Grand Strategy: Islamist Militancy, National Security, and the Pakistani State*. Oxford: Oxford University Press, 2017.

As such, Pakistan fostered a “military-madrassa-mullah” complex that has utilized a militant brand of Islam to defend the state against its “existential enemies.”⁹² Nationalism was packaged as religiosity. Such ideological links have largely continued; with India’s increasing aggression—and Hindu nationalist character—its role as one of Pakistan’s ‘existential enemies’ has only been amplified. As South Asia became shadowed with the possibility, and soon the actualization, of nuclear weapons, Islamist proxies became increasingly important. Indeed, as Kapur argues, Islamist proxies could not have lasted as long as they have “in a nonnuclear environment.”⁹³ Pakistani literature on the country’s strategic approach to India repeatedly references how nuclear weapons can increase impunity for its support of non-state actors, allowing Pakistan to revisit the Kashmir issue without reliving 1971.⁹⁴ Certainly, until India responded with an air strike in 2019, Pakistani-sponsored terrorist attacks went almost always unanswered with India’s conventional might. Islamabad successfully trapped New Delhi into fearing its ambiguous nuclear threshold.

Pakistan’s Nuclear Doctrine—Shifting to Asymmetric Escalation

Since its acquisition of nuclear weapons, Pakistan’s nuclear doctrine has shifted away from a catalytic nuclear posture to an asymmetric escalation posture. Asymmetric escalation involves the rapid first use of nuclear weapons against conventional forces—particularly when the balance of power favors the adversary.⁹⁵ Instead of leveraging the nation’s benefactor state, the U.S., Pakistan has leveraged nuclear weapons to achieve strikingly higher levels of deterrence than India despite conventional inferiority. However, there are inherent risks to such a posture: Pakistan requires a credible first-use posture and therefore sacrifices a “substantial degree of assertive and centralized control over its nuclear assets.”⁹⁶ It must be able to react quickly in a crisis—particularly in the case of an Indian invasion—which implies that there must be some degree of delegative control. As Peter Feaver argues in “Command and Control in Emerging Nuclear Nations,” the “greater the fear of coups, the more likely the [nuclear] weapons are under assertive control.”⁹⁷ Pakistan must reckon with the command and control pressures brought on both by its need for credible first use and its political instability. In a nation that faces frequent military intervention and unstable civil-military relations, the delegative control needed for secure nuclear handling during a crisis is not necessarily secure. Therefore, Pakistan must constantly reckon with this “vulnerability/invulnerability paradox” of balancing deterrence with weapons management should it continue to try and leverage nuclear weapons to sub-conventionally challenge India.⁹⁸ As New Delhi begins to call Pakistan’s nuclear bluff, the nation’s delegative control may be further challenged.

⁹² Mohanty, Sanjeeb Kumar, and Jinendra Nath Mahanty. “Military-Madrassa-Mullah Complex: Promoting Jihadist Islam in Pakistan.” *India Quarterly* 66, no. 2 (2010): 133–49. <http://www.jstor.org/stable/45401687>.

⁹³ Kapur, *Jihad as Grand Strategy*, 2017.

⁹⁴ Fair, C. Christine. “Jihad under the Nuclear Umbrella.” *Fighting to the End*, May 27, 2014, 226–60. <https://doi.org/10.1093/acprof:oso/9780199892709.003.0009>.

⁹⁵ Narang, *Posturing for Peace?*, 2022.

⁹⁶ Ibid.

⁹⁷ Feaver, Peter D. “Command and Control in Emerging Nuclear Nations.” *International Security* 17, no. 3 (1992): 160–187. <https://muse.jhu.edu/article/447036>.

⁹⁸ Narang, *Posturing for Peace?*, 2022.

Catalytic nuclear postures, instead, relies on centralized command and control; by leveraging ambiguous nuclear capabilities, a state aims to catalyze a third party, usually the U.S. or other benefactor states, into intervening on its behalf.⁹⁹ Indeed, Pakistan's changes in nuclear posture is largely the result of the U.S.'s weakening support. In 1989-1990, when faced with internal dissent in Indian-administered Kashmir, India quickly blamed Pakistan. The Indian army thereby decided to mobilize its forces towards northern India and deploy its strike corps; Pakistan, in turn, countermobilized, but the balance of conventional military favored India's forces.¹⁰⁰ Pakistan deliberately signaled to the United States—and not directly to India—that it was preparing to use its nuclear capabilities, intending to trigger US intervention. The U.S. then played a critical role in defusing the crises before nuclear weapons could be involved.¹⁰¹ Whether Pakistan truly was ready to use its nuclear weapons was unclear, but the intervention of the U.S. was undoubtedly pivotal in reducing tension. While in this case, Pakistan did not necessarily deter India, it definitely catalyzed a U.S. response. This posture was once particularly optimal for a Pakistan that wished to prevent India from becoming overtly nuclear itself, a gamble that relied on overwhelming U.S. strategic interests in Pakistan. After India's 1998 nuclear tests and the Kargil crisis, India became an established nuclear power, and Pakistan visibly lost the support of the U.S. during an India-Pakistan conflict. In reaction, Islamabad developed an asymmetric escalation posture to pursue its strategic interests without facing conventional threats to its existential statehood. The technologies and strategies Pakistan has developed over the last two decades reflect this change in posture.

Full Spectrum Deterrence and Non-Strategic Options

In 2013, Pakistan's National Command Authority announced that Pakistan would be pursuing "credible minimum full spectrum deterrence."¹⁰² While the necessities of full spectrum deterrence (FSD) is not entirely clear, Pakistani press releases imply that it entails deterring India in every possible theatre while stockpiling a substantial number of diverse nuclear options.¹⁰³ Pakistan intends on having nuclear capabilities at all ranges and all yields. FSD is partially a response to the Cold Start. Cold Start is essentially a strategy of limited conventional war; rather than a full-scale strategic strike, through tactical operations, India aimed to conventionally confront Pakistan. Initially, Cold Start may have threatened Pakistan's credible deterrence. In 2001, retired Lt. Gen Khalid Kidwai had announced that Pakistan's nuclear weapons were entirely aimed at deterring nuclear aggression that posed an existential threat.¹⁰⁴ He specified that Pakistan would resort to nuclear weapons if India crossed a space, military, economic, or domestic threshold that completely threatened Pakistan's survival. In contrast, FSD reinforces

⁹⁹ Ibid.

¹⁰⁰ Zeb, Rizwan. *Revisiting the Role of Nuclear Weapons in India-Pakistan Conflict*. Islamabad: Institute of Strategic Studies Islamabad, 2021. https://issi.org.pk/wp-content/uploads/2021/05/1_SS_Rizwan-Zeb_No-1_2021.pdf.

¹⁰¹ Narang, *Posturing for Peace?*, 2022.

¹⁰² Khan, Moiz. *Understanding Pakistan's Full Spectrum Deterrence*. Islamabad: Institute of Strategic Studies Islamabad, 2020. https://issi.org.pk/wp-content/uploads/2020/11/1_SS_Moiz-Khan_No-6_2020.pdf.

¹⁰³ Ibid.

¹⁰⁴ Pugwash Conferences on Science and World Affairs. *Report on Nuclear Safety, Nuclear Stability, and Nuclear Strategy in Pakistan*. January 14, 2002. <https://pugwash.org/2002/01/14/report-on-nuclear-safety-nuclear-stability-and-nuclear-strategy-in-pakistan/>.

Pakistan's credible minimum deterrence. As described by General Kidwai, a member of Pakistan's National Command Authority, FSD provides Pakistan the "liberty to select from a range of countervalue, battlefield, and counterforce targets."¹⁰⁵ By developing a diverse range of options, Pakistan makes itself unpredictable while attempting to deny India an advantage in any theatre. As Cold Start hinged on operating below Pakistan's nuclear threshold, FSD allows Pakistan to react in a range of scenarios, effectively clouding and lowering its nuclear red line.

Most notably, in Pakistan's quest for FSD, it has concentrated investment in short-range and cruise missile capabilities. Strategically, Pakistan possesses credible minimum deterrence with its range of medium and intermediate range weapons. Pakistan owns the Ghauri (Hatf 5) medium-range, road-mobile, liquid-fueled ballistic missile with a range of 1,500 km.¹⁰⁶ Capable of carrying a 12-35 kT nuclear warhead, the Ghauri missile can threaten targets deep inside Indian territory. The first test flight took place in 1998 and the missile entered Pakistan's service in 2003. Furthermore, Pakistan developed the two-stage solid-fueled road mobile Shaheen II (Hatf 6) with a range of 1500-2000 km, a medium-range ballistic missile capable of reaching major cities in India.¹⁰⁷ The Shaheen III, a two-stage, solid-fueled, road mobile missile, with a reported range of 2,750 km, is currently in development. It has been tested at least 3 times (2015, 2021, and 2022) and was publicly displayed in a 2016 military parade. However, it has not been deployed. With the addition of the Shaheen III, Pakistan can confidently hold nearly every major Indian city at risk. While the Shaheen III's recent tests were potentially aimed at countering India's aggressive modernization, Pakistan has long been capable of carrying out a countervalue strike.¹⁰⁸

However, its deterrence against limited conventional strikes would need to come from shorter range missiles. Tactical nuclear weapons (TNWs) could be wielded on the battlefield to immediately offset India's conventional superiority. Indeed, the introduction of the Nasr (Hatf 9) missile in 2011 as a "quick response" delivery system¹⁰⁹ is designed to meet "evolving threats like Cold Start."¹¹⁰ General Qamar Javed Bajwa reportedly claimed that "Nasr has put cold water on Cold Start."¹¹¹ Employing short-range ballistic missiles helped plug gaps at the operational and tactical levels. The Nasr missile is a short-range surface to surface ballistic

¹⁰⁵ Gul, Imran. "Pakistan's Full Spectrum Deterrence: Trends and Trajectories." *South Asian Voices*, August 22, 2023. <https://southasianvoices.org/pakistan-full-spectrum-deterrence-trends-trajectories/>.

¹⁰⁶ "Hatf-5." *Missile Threat*, Center for Strategic and International Studies (CSIS). Accessed August 11, 2024. <https://missilethreat.csis.org/missile/hatf-5/>.

¹⁰⁷ "Hatf-6." *Missile Threat*, Center for Strategic and International Studies (CSIS). Accessed August 11, 2024. <https://missilethreat.csis.org/missile/hatf-6/>.

¹⁰⁸ Centre for Strategic and Contemporary Research (CSCR). "Pakistan's Test of Shaheen-III: Countering Indian Aggression." *Centre for Strategic and Contemporary Research*. Accessed August 11, 2024. <https://cscr.pk/explore/themes/defense-security/pakistans-test-of-shaheen-iii-countering-indian-aggression/>.

¹⁰⁹ "Hatf-9." *Missile Threat*, Center for Strategic and International Studies (CSIS). Accessed August 11, 2024. <https://missilethreat.csis.org/missile/hatf-9/>.

¹¹⁰ Khan, Shuja. "Pakistan's Tactical Nuclear Weapons and Their Impact on Stability." *Carnegie Endowment for International Peace*, June 21, 2016. <https://carnegieendowment.org/research/2016/06/pakistans-tactical-nuclear-weapons-and-their-impact-on-stability?lang=en>.

¹¹¹ Rajagopalan, Rajeswari Pillai. "Pakistan's Nasr Missile: 'cold Water' over India's 'Cold Start'?" *orfonline.org*, December 4, 2023. <https://www.orfonline.org/research/pakistans-nasr-missile-cold-water-over-indias-cold-start-47711#:~:text=After%20a%20previous%20test%20launch,firing%20on%2019%20April%202011>.

missile with a range of 60-70 kilometers. It can carry a single warhead of around 400 kg and can be launched with enough accuracy to be wielded in battlefield scenarios.¹¹² Before the Nasr, Pakistan could have utilized the Babur (Hatf 7) ground-launched cruise missile, constructed in response to India's burgeoning cruise missile program in the 1990s, which could deliver a warhead to a range of 350-700 kms. It also held the 100-km Shaheen 1/1A, the 180-km Abdali, and the 280-km Ghaznavi. While this is quite large in comparison to the Nasr, it still could potentially challenge operational and tactical movements by Indian forces.¹¹³

Even if Pakistan has TNWs, the introduction of nuclear weapons into a conventional scenario requires leaders to consider several potential causes. Critics of Pakistan's TNW program have referred to the vulnerability/invulnerability paradox inherent in the use of tactical weapons. In order to be effective as counterforce, weapons like the Nasr would need to be deployed early, forcing Pakistani decision-makers to react quickly to avoid needing to escalate to strategic weapons. While Pakistani officials have claimed they would maintain assertive control over battlefield weapons, centralized control may be difficult when maintaining efficiency, especially as it is partially kept through in-built redundancies in the command and control structure.¹¹⁴ It is possible that Pakistan can still preserve tactical credible minimal deterrence while retaining assertive control; however, the lack of detail about what this system might be remains concerning.

Nonetheless, Pakistan does possess the Pakistani-built Permissive Action Link, a twelve-digit alphanumeric code, that prevents unauthorized launch and is built into nuclear weapon technology during manufacture.¹¹⁵ Launch codes further follow a two-man rule where codes are held by the NCA and fed to local commanders to authorize a launch. Pakistani officials claim to have guarded against potential communication failures through multiple redundancies and backup channels.¹¹⁶ As such, the operation of a two-man system under complete communication failure is unlikely. Furthermore, there are several Indian invasion scenarios that may force Pakistan to use TNWs in its own territory. The Nasr has a 60-km range. In a shallow invasive thrust—as envisaged by Cold Start—if Indian forces penetrated far enough into Pakistan, decision-makers would need to consider employing the Nasr in Pakistani territory. In a Cold Start invasion strategy, some of India's armored contingents are likely to be deployed alongside the highly prized Punjab border. Deploying a nuclear weapon on one's own soil would rapidly escalate the conflict while threatening political good will with the Pakistani people.

Pakistan's TNW program remains a necessity to truly attain FSD according to multiple military and civilian leaders. General Kidwai describes the Nasr as a force multiplier that enhances Pakistan's deterrent capabilities across the threat spectrum. He argues that such weapons make war less likely by preventing India from limited conventional war. Furthermore, the Nasr allows decision-makers to respond proportionately without being forced into a similar

¹¹² Khan, *Understanding Pakistan's Full Spectrum Deterrence*, 2020.

¹¹³ Ibid.

¹¹⁴ Khan, *Pakistan's Tactical Nuclear Weapons*, 2016.

¹¹⁵ Luongo, Michael. "Pakistan's Nuclear Policy and the Future of South Asian Stability." *Arms Control Today*, December 2007. https://www.armscontrol.org/act/2007_12/Luongo.

¹¹⁶ Khan, *Pakistan's Tactical Nuclear Weapons*, 2016.

position India has been in—either forced to not respond to an attack or respond and risk strategic-scale nuclear war. Military leaders maintain that FSD is deterrence-focused and not a warfighting strategy, presenting TNWs as an extension of existing conventional and nuclear doctrine.¹¹⁷ Nonetheless, as Balakot demonstrated, there are avenues to engage Pakistan in conflict without necessarily triggering the use of TNWs. Furthermore, as India strengthens its ability to retaliate with both low-yield and high-yield weapons and reinforces its posture of assured retaliation, the deployment of Nasr, even to prevent an Indian invasion, could be disastrous. While FSD may be deterrence-focused now, the capabilities Pakistan is developing would inevitably be used in warfighting should deterrence fail. If India can truly realize a counterforce strike against the Nasr, Pakistan would need to use it or lose the ability to launch it entirely.

Second Strike Survivability—MIRV Technology and the Naval Theatre

Vital to any nuclear state is the survivability of their forces. As previously discussed, Pakistan may logically perceive India's military modernization as preparing for counterforce targeting. Therefore, Pakistan must also take steps to protect the longevity of its arsenal and ensure that it can still retaliate even with weakened second strike capabilities. To this end, Pakistan is working on MIRV technology while investing in the sea-leg of its nuclear triad. Both ventures, however, have not been without their challenges. There have been doubts about the functionality of Pakistan's MIRVed Ababeel missile while its burgeoning nuclear submarine program is rife with obstacles and questions.

The Ababeel is Pakistan's first surface-to-surface, three-stage, solid-fueled missile ballistic missile with an estimated range of 2,200km. First unveiled in 2017, the Ababeel reportedly contains MIRV technology that Pakistan developed with help from Chinese engineers. However, during the Ababeel's first flight test in 2017, the missile fell quite short of what it was reportedly able to reach—at most 1100 km according to one analyst. Furthermore, it is unclear whether the MIRV technology that Ababeel supposedly wields was employed during the failed test.¹¹⁸ Pakistan again tested the weapon in 2023, this time successfully.¹¹⁹ However, some experts claim that Pakistan's MIRV success is unverified.¹²⁰ Until the Ababeel enters the service and is operationally deployed, there may still be lingering doubts. The Ababeel falls under Pakistan's FSD doctrine of denying India's advantage even in a potential countervalue strike. The Ababeel is a response to India's quest for robust BMD—to combat a robust BMD system MIRV technology supports Pakistan's second strike capacity by increasing the probability that nuclear warheads can reach Indian targets. Paradoxically, the successful test levels the mutual vulnerability between India and Pakistan. India has tested its own MIRVed Agni-P, implying that further evolution in similar technologies is likely to continue in South Asia. Whether or not

¹¹⁷ Ibid.

¹¹⁸ Center for Strategic and International Studies. "Ababeel." Missile Threat. Accessed August 12, 2024. <https://missilethreat.csis.org/missile/ababeel/>.

¹¹⁹ Sarkar, Aditya. "Why Did Pakistan Test Its MIRV-Capable Ababeel Missile?" *The Diplomat*, November 20, 2023. <https://thediplomat.com/2023/11/why-did-pakistan-test-its-mirv-capable-ababeel-missile/>.

¹²⁰ London, Brad, and Rhea Mogul. "India Joins Select Group of Nations Able to Fire Multiple Warheads on a Single ICBM." CNN, March 12, 2024. <https://www.cnn.com/2024/03/12/india/india-mirv-icbm-intl-hnk-ml/index.html>.

Ababeel can survive Indian BMD and the MIRV technology can be effectively deployed, however, is unclear. Pakistani decision-makers can nonetheless be comforted by the fact that India's BMD has also not been put to the test against the type of missile it claims to have. Until Pakistan can prove it has reached the level of sophistication and confidence India has in its Agni-P, however, there will remain some degree of strategic imbalance.¹²¹

A successfully deployed Ababeel may decrease asymmetry between India and Pakistan, lending some stability, Pakistan's other second-strike ventures face significant logistical and security challenges. True survivability for Pakistan would arise from naval nuclear capabilities. General Kidwai acknowledged that land-based second strike capabilities are not necessarily assured.¹²² As India attains conventional and nuclear capabilities that could cripple some portion of Pakistan's arsenal, diversifying its nuclear weapons would allow Pakistan to ensure at least some forces survive.

In January 2017, Pakistan tested its Babur-3, a submarine-launched cruise missile with a reported range of 450 km. The Pakistani military claimed that Babur-3 is "capable of delivering various types of payloads and will provide Pakistan with a Credible Second Strike Capability, augmenting deterrence."¹²³ Such an addition to Pakistan's arsenal may seem optimistically stabilizing. Assured second strike capabilities would disincentive India from launching a counterforce strike. Sea-based weapons may also seem to decrease "use-it-or-lose it" pressures that may arise from fears of early nuclear escalation. However, as Ankit Panda and Christopher Clary argue in "Safer at Sea? Pakistan's Sea-Based Deterrent and Nuclear Weapons Security," Pakistan's naval weapons "bring with them new dangers, and their basing mode at sea does not eliminate many dangers that sea-based weapons share with land-based forces."¹²⁴ Not only does a sea-based nuclear force raise new questions of command and control, but threats to Pakistan's naval bases indicate that its nuclear naval arsenal face similar threats from adversarial forces as land-based forces. Furthermore, as Pakistan-China cooperation dramatically improves Pakistan's conventional naval capabilities, India will need to react to maintain its strategic prowess in the naval arena.¹²⁵

Pakistan currently does not intend to conduct deterrent patrols with its submarines. Instead, its submarines would be flushed at the onset of a crisis while remaining berthed during peacetime. While this may decrease concerns about nuclear weapons being separated from central command, Pakistani submarines' peacetime locations are not secret. Unlike aircraft or

¹²¹ Zohaib Altaf, Nimrah Javed. "The Triad of Technology and Its Implications for Strategic Stability in South Asia." *South Asian Voices*, June 3, 2024. <https://southasianvoices.org/sec-c-pk-r-triad-of-technology-05-02-2024/>.

¹²² Clary, Christopher, and Ankit Panda. 2017. "Safer at Sea? Pakistan's Sea-Based Deterrent and Nuclear Weapons Security." *The Washington Quarterly* 40 (3): 149–68. doi:10.1080/0163660X.2017.1370344.

¹²³ "Pakistan Conducts First Test of Submarine-Launched Babur-3 Cruise Missile." *Missile Threat: CSIS Missile Defense Project*, January 9, 2017. <https://missilethreat.csis.org/pakistan-conducts-first-test-submarine-launched-babur-3-cruise-missile/>.

¹²⁴ Clary, Christopher, and Ankit Panda. "Safer at Sea? Pakistan's Sea-Based Deterrent and Nuclear Weapons Security." *The Washington Quarterly* 40, no. 3 (2017): 149–68.

¹²⁵ Kathju, Junaid. "Underwater Race: Pakistan's Chinese Submarines Set to Spur Indian Navy's Upgrade." *South China Morning Post*, May 3, 2024. <https://www.scmp.com/week-asia/politics/article/3261325/pakistans-chinese-stealth-submarines-set-spur-indian-navys-upgrade-beijing-eyes-oceanic-expansion>.

land-based weapons that can be operated from a plethora of locations, Pakistani leaders have only two options to store nuclear-capable submarines. The Karachi or Jinnah shipyard where these vehicles can be stored are highly visible and can be targeted by Indian conventional and nuclear forces.¹²⁶ In a crisis situation, particularly during conventional war, India would be tempted to target their naval capabilities before Pakistan can send them out to sea. Conversely, Pakistan would be tempted to send them out as quickly as possible to avoid a potential Indian counterforce strike.

Should Pakistan choose to begin deterrence patrols, it will need to contend with India's anti-submarine warfare capabilities. Alongside India's ISR capabilities, it has been working to improve its anti-submarine missiles and conventional naval might. As the Babur-3 has a 450 km range, shorter than the distance from Karachi to India's closest cities, India will aim to at the least restrain the submarine to Pakistan's littoral waters to prevent an attack on its citizenry by eliminating any submarine it can.¹²⁷ Furthermore, even outside of an all-out war scenario, if Pakistan does not signal whether its submarines are strategic or conventional, there is a risk of inadvertent escalation. As Panda and Clarey note, "it would be difficult if not impossible for an Indian Navy surface ship, submarine, or maritime reconnaissance aircraft to know if a detected Pakistani submarine has a strategic or conventional mission."¹²⁸ If the Indian Navy encounters a Pakistani submarine, it will try to destroy it, increasing use-it-or-lose-it pressures if leaders believe Pakistani submarines are being targeted. Nuclear-capable submarines increase credible minimum deterrence as they are substantially invulnerable to an adversary's anti-submarine capabilities. India's desire to augment its anti-submarine arsenal can be read as a reaction to ensure Pakistani submarines remain vulnerable in Indian waters.

India may miss the window to eliminate Pakistan's submarines before they are flushed to sea; in this situation, India can try to cripple Pakistan's line of communication between the National Command Authority and the submarine's crew. Pakistan communicates to its submarines through very low frequency (VLF) communication. In 2016, Pakistan revealed its VLF facility at its naval base PNS Hameed near the Karachi dockyard.¹²⁹ If India either cannot target the submarines prior to their launch or wishes to not strike a Pakistani port, it can choose to target this VLF facility. Submarines would then need to surface to receive signals, increasing their vulnerability, or risk becoming futile should on-board commanders not have the authority to order a launch themselves. As Pakistani leaders have repeatedly asserted their intent to continue centralized command even with its naval nuclear forces, such delegative authority may not exist. Indeed, Pakistani assurances about centralized command raise questions about how nuclear submarines can be utilized in a crisis. Without a delegative command structure, such submarines

¹²⁶ Panda, Ankit. "The Risks of Pakistan's Sea-Based Nuclear Weapons." – *The Diplomat*, October 17, 2017. <https://thediplomat.com/2017/10/the-risks-of-pakistans-sea-based-nuclear-weapons/>.

¹²⁷ Clary, Christopher, and Ankit Panda. "Safer at Sea? Pakistan's Sea-Based Deterrent and Nuclear Weapons Security." *The Washington Quarterly* 40, no. 3 (2017): 149–68.

¹²⁸ Ibid.

¹²⁹ Ansari, Usman. "Pakistan Unveils VLF Submarine Communications Facility." *Defense News*, August 19, 2022. <https://www.defensenews.com/naval/2016/11/16/pakistan-unveils-vlf-submarine-communications-facility/>.

are incredibly vulnerable to communication failure; on the other hand, allowing for delegative command and control would only increase fears of unauthorized use for both Pakistani leaders and international observers.

Part of Pakistan's challenges in ensuring credible minimum deterrence is the limited size of its nuclear naval arsenal. With only 5 submarines, India can ostensibly target its entire force when docked and attempt to track and destroy each submarine should they be flushed out. However, the Pakistan's recent acquisition of Chinese vehicles would add 8 more nuclear-capable attack submarines to its arsenal. Furthermore, the Hangor-class submarine's stealth capabilities are superior to both the Hashmat and Khalid submarines. In fact, the submarines boast advanced technology such as air-independent propulsion, which increases the amount of time a submarine can remain submerged, and new underwater sensors that edge out India's best Kalvari class submarines.¹³⁰ While India does have nuclear-powered ballistic submarines, such as the INS Arihant, it will be discomfited by Pakistan's improved dual-capable submarine arsenal. In reaction to Pakistan's acquisition of Chinese submarines, New Delhi will likely amp up both its anti-submarine capabilities and its submarine technology. The first of eight Hangor submarine was launched in April 2024. Yet, Pakistan has remained vague about whether they will take a nuclear role. Nonetheless, as Pakistan's capabilities grow, India will need to continue to find ways to keep Pakistani submarines vulnerable.

Conclusion

South Asia's tightening arms race is harming the region's stability. India and Pakistan are no longer newborn nuclear states content with minimal warheads and little strategic depth. Nor is India still satisfied with being trapped by Pakistan's nuclear shadow. As India dramatically modernizes its military, Pakistani leaders are wary of the changes in targeting and operating doctrine that New Delhi may be willing to make. As Pakistan continues its support for cross-border terrorism, India will look to find ways to respond and inflict retaliatory damages. Full Spectrum Deterrence and the Nasr missile were reactions to India's Cold Start plans. Cold start was a reaction to Pakistan's continual threat of asymmetric escalation. Caught in an action-reaction dynamic, each state must continue to proliferate their nuclear, conventional, and ISR capabilities or risk becoming overwhelmingly disadvantaged in a crisis. It is true that nuclear deterrence in South Asia remains stable. The advancements discussed in this paper occurred over the span of nearly 20 years, and mutual vulnerability has yet to be truly broken. Leaders from both states are incredibly aware of how closely the adversary is watching—as such, they act cautiously during peacetime to try and prevent inadvertently signaling the intent to escalate. During crises, Islamabad and New Delhi have found ways to deescalate before following through on their nuclear threats.

However, the types of weapons India and Pakistan now wield are far beyond anything India and Pakistan have used before. If New Delhi and Islamabad cannot adapt their

¹³⁰ Kathju, Junaid. "Underwater Race: Pakistan's Chinese Submarines Set to Spur Indian Navy's Upgrade." *South China Morning Post*, May 3, 2024.

command and control structures to ensure safe, responsible, but effective maintenance of their weaponry, then escalation is far more likely. With China's extensive nuclear strength challenging both India's deterrence capabilities and global prestige, India is not likely to reduce its military modernization. If Pakistan wishes to maintain credible deterrence against its primary adversary, it cannot stop advancing its own arsenal. As such, the current and future military technological shift in India and Pakistan have escalated the risk of nuclear conflict by increasing crisis instability and creating avenues for limited nuclear engagements. Moreover, as India and Pakistan tackle new technologies, there are risks that their command and control structures have not yet adapted to ensure such capabilities can be used credibly, blunting any supposed improvement in deterrence advanced technology may bring.

Modi's choice to send air forces across the Line of Control in 2019 signals serious intent to retaliate when provoked. While India still maintains a 'No First Use' posture, its increasing hawkishness and military modernization undoubtedly indicates India's intensifying counterforce capabilities. Currently, India cannot carry-out a full-scale counterforce campaign. Yet, India's missiles are more accurate and its warheads more ready. It is developing improved enabling technologies—particularly ISR and BMD—that actualizes types of targeting today that were impossible a decade ago. As such, if Pakistan wishes to continue counterbalancing India's superior conventional forces, it will need to act before India can decimate any substantial portion of its nuclear capabilities. In a crisis, such 'use-it-or-lose-it' pressures are deeply destabilizing, especially when recent conflicts have been resolved mostly through chance. Furthermore, India's ambiguity about their targeting doctrine will force Pakistan to consider worst-case scenarios when deciding their course of action. As many of Pakistan's scenarios for nuclear use arise from conventional war, India's newfound willingness to risk conventional conflicts and burgeoning ability to preempt Pakistani nuclear attacks is discomfiting.

Similarly, Pakistan's adoption of Full Spectrum Deterrence is both a reaction to Cold Start and India's potential counterforce capabilities. In order to achieve FSD, Pakistan invested in missiles of all ranges that can be used from the battlefield to a strategic attack. Furthermore, to bolster its second strike survivability, Pakistan has invested in nuclear naval forces that can be flushed out during a crisis. However, if Pakistan intends to maintain assertive control rather than delegate downwards, it will need to consider how such weapons can effectively be used in a crisis scenario. Tactical nuclear weapons may be enough to deter a conventional assault, but only if Pakistan can use them in a timely manner. If not, Pakistan would need to escalate to far more costly options, elevating the conflict beyond a limited engagement. Nuclear submarines face obstacles as well. Easily visible when berthed, India would be tempted to preemptively strike Pakistan's submarines. While in the water, India has no reason not to destroy any submarine they can find. If Pakistan wants its naval deterrent to be credible, it will need to continuously improve—as it has through cooperation with China—to deny India any advantage in the ocean. Pakistan's forces may also face attacks from non-state actors within the state who seek to leverage nuclear weapons for their own purposes. Non-state Islamist actors

exploit debates on nuclear policy to increase their political influence¹³¹ and frequently make statements about buying weaponry.¹³² In the context of Pakistan's existing military-madrassa-mullah complex that binds religious extremism to Pakistani military and intelligence, concerns about such attacks will undoubtedly raise alarms in both India and the U.S.¹³³

If India and Pakistan wish to reduce regional tensions, then both states should seriously consider introducing confidence-building measures and arms control to slow down the speed of their arms race. Bilateral arms agreements, however, are complicated by India's desire to attain global prestige and compete with the far-superior forces of China at its Eastern border. Indo-Chinese arms agreements are similarly difficult as China seeks to compete with the U.S. Moreover, India is loathe to negotiate with a Pakistan that it perceives as tacitly supporting terrorist groups. On its end, Pakistani leaders seem concerned with the increasing Hindu nationalist character of New Delhi's BJP government. Therefore, the action-reaction dynamics of the South Asian arms race is likely to continue—and even spiral faster—as India and Pakistan invest as much as they can into arsenals far larger than what credible minimum deterrence requires. Now, however, India seems willing to call Pakistan's bluff.

¹³¹ Joshi, Sharad. 2019. "How Terrorist Actors in Pakistan Use Nuclear Weapons for Political Influence." *Asian Security* 16 (2): 221–42. doi:10.1080/14799855.2019.1582522.

¹³² "Who control's Pakistan's atomic bomb? Hafiz Saeed says it's 'asset of Islam', use it for jihad against US," *Financial Express*, January 2, 2018.